

# Retail Sales Data Analysis

## Major Project Documentation

An end-to-end data analytics project covering data cleaning, preprocessing, feature engineering, exploratory data analysis, and visualization of retail sales transactions using Python (Pandas, NumPy, Matplotlib).

| Tool / Environment     | Jupyter Notebook (Python 3.13) |
|------------------------|--------------------------------|
| Core Libraries         | pandas, numpy, matplotlib      |
| Dataset                | retail_sales_raw.xlsx          |
| Final Records Analyzed | 1,002 transactions             |

# 1. Project Overview

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This project performs a complete analysis of a retail sales dataset, taking it from a raw, inconsistent Excel export through cleaning, validation, feature engineering, and finally exploratory visualization to surface business insights such as revenue trends, top-performing products, and regional performance.

## 1.1 Objectives

- Clean and standardize a messy real-world retail sales dataset.
- Handle missing values, duplicate records, and inconsistent formatting.
- Engineer time-based and derived business features (Year, Quarter, Month, Gross Amount, etc.).
- Compute key business metrics (Total Revenue, Total Orders, Average Order Value).
- Visualize revenue trends across time, category, region, and product lines.

## 1.2 Tools & Technologies

| Component               | Technology         |
|-------------------------|--------------------|
| Language                | Python 3.13        |
| Development Environment | Jupyter Notebook   |
| Data Handling           | pandas, numpy      |
| Visualization           | matplotlib         |
| File Formats            | Excel (.xlsx), CSV |

## 2. Dataset Description

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The raw dataset (**retail\_sales\_raw.xlsx**) contains 1,020 retail transaction records across 13 columns, capturing order, customer, product, and payment details.

| Column        | Description                                        |
|---------------|----------------------------------------------------|
| OrderID       | Unique identifier for each order                   |
| OrderDate     | Date the order was placed                          |
| CustomerName  | Name of the customer                               |
| Category      | Product category (e.g. Sports, Electronics, Books) |
| Product       | Specific product name                              |
| Region        | Sales region (North / South / East / West)         |
| City          | City of the transaction                            |
| Quantity      | Units purchased                                    |
| UnitPrice     | Price per unit                                     |
| DiscountPct   | Discount applied (%)                               |
| Revenue       | Net revenue generated                              |
| PaymentMethod | Mode of payment used                               |
| SalesRep      | Sales representative handling the order            |

Initial inspection (`df.shape`, `df.info()`) revealed 1,020 rows and multiple data quality issues, including missing values, inconsistent text casing, mixed date formats, and price fields stored as text with currency symbols.

## 3. Data Cleaning & Preprocessing

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The raw data required substantial cleaning before analysis. The key steps performed, in order, were:

### 3.1 Missing Value Handling

- **CustomerName**: 31 missing values filled with the placeholder 'Unknown'.
- **City, Quantity, UnitPrice, PaymentMethod**: rows retained; nulls addressed during downstream numeric conversion and analysis.
- **DiscountPct**: 30 missing values imputed using the column median to avoid skewing revenue calculations.

### 3.2 Duplicate Removal

Duplicate rows were identified using `df.duplicated()` and removed with `drop_duplicates()`, reducing the working dataset from 1,020 to 1,002 clean records.

### 3.3 Text Standardization

- Stripped leading/trailing whitespace from column headers and text fields.
- Applied consistent **Title Case** formatting to CustomerName, Category, Product, Region, City, PaymentMethod, and SalesRep (e.g. 'SPORTS' / 'Sports' → 'Sports').
- This resolved inconsistent category labels caused by mixed casing in the source file.

### 3.4 Date Parsing

OrderDate contained mixed date formats (e.g. DD/MM/YYYY and MM/DD/YYYY). This was resolved using pandas' flexible parser with `format="mixed"` and `errors="coerce"` to safely convert all values to proper datetime objects.

### 3.5 Currency & Numeric Cleanup

UnitPrice values contained currency symbols (■, \$) and thousands separators, causing the column to be stored as text. These characters were stripped and the column was converted to numeric type using `pd.to_numeric(errors="coerce")`.

### 3.6 Validity Checks

- Checked for invalid records where  $\text{Quantity} \leq 0$ ,  $\text{UnitPrice} \leq 0$ ,  $\text{Revenue} < 0$ , or  $\text{DiscountPct} > 100\%$ , to flag potential data entry errors.
- Verified category, region, and payment method values were limited to a clean, consistent set after standardization.

## 4. Feature Engineering

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New columns were derived from OrderDate and existing fields to support time-based and financial analysis:

| New Column      | Derivation                                            |
|-----------------|-------------------------------------------------------|
| Year            | Extracted from OrderDate                              |
| MonthNo / Month | Numeric and named month extracted from OrderDate      |
| Quarter         | Calendar quarter (Q1-Q4) derived from OrderDate       |
| Day             | Day of week name                                      |
| Week            | ISO calendar week number                              |
| GrossAmount     | Quantity x UnitPrice (pre-discount transaction value) |

The cleaned and enriched dataset was exported to **Retail\_Sales\_Cleaned.csv** and **Retail\_Sales\_Cleaned.xlsx** for reuse in downstream analysis and reporting (e.g. Power BI).

## 5. Key Business Metrics

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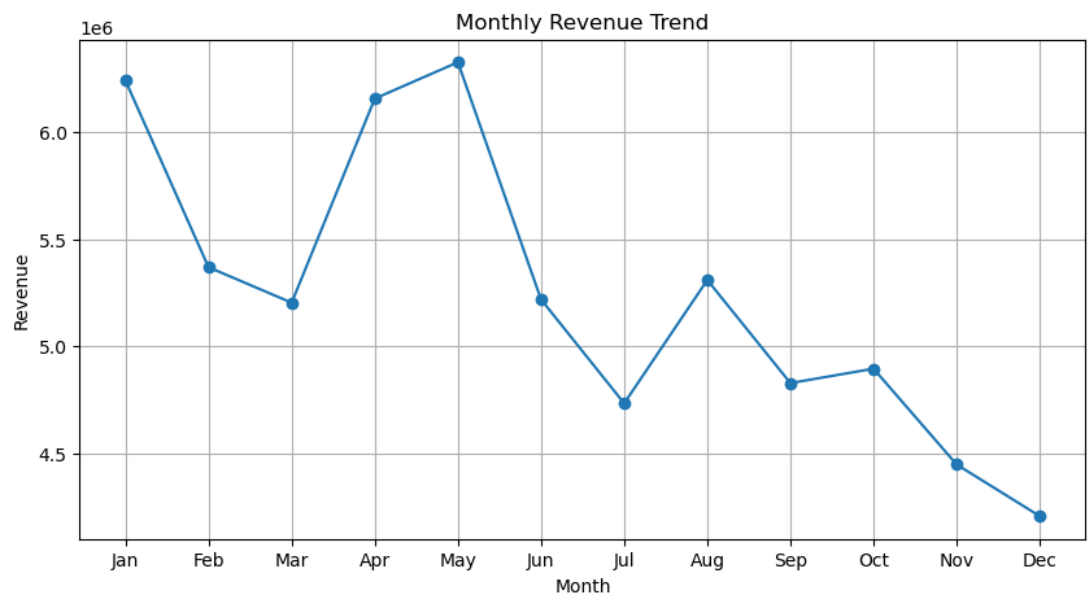
After cleaning, the following headline metrics were computed across all 1,002 transactions:

| Metric                     | Value              |
|----------------------------|--------------------|
| Total Revenue              | Rs. 6,29,28,075.72 |
| Total Orders               | 1,002              |
| Total Quantity Sold        | 5,402 units        |
| Gross Sales (pre-discount) | Rs. 6,58,26,336.29 |
| Average Order Value        | Rs. 62,802.47      |

## 6. Exploratory Data Analysis & Visualizations

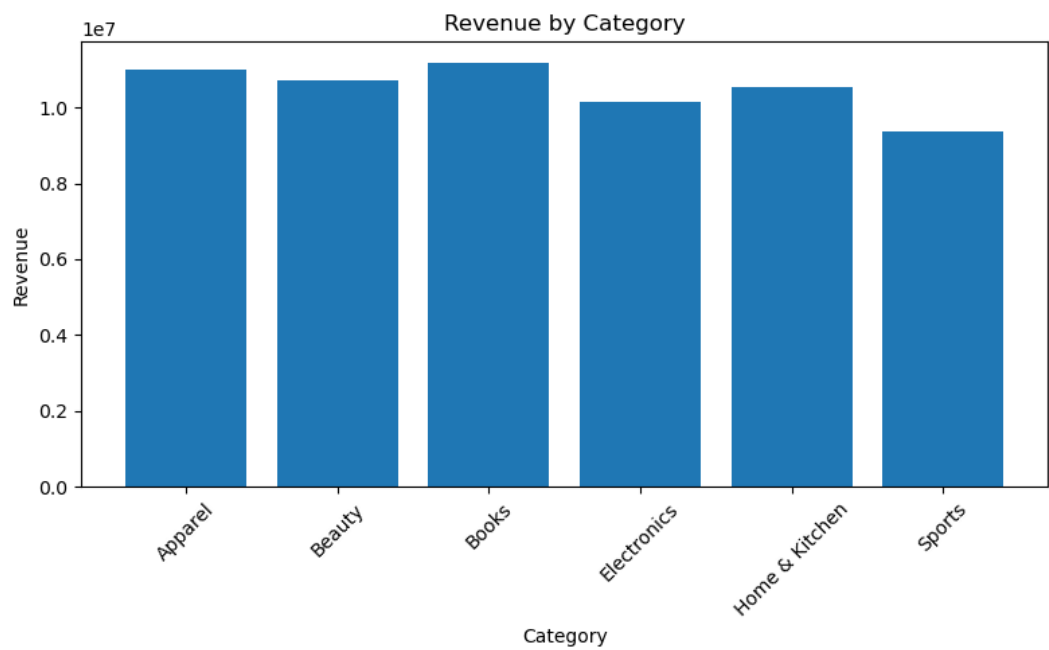
The following charts were generated with matplotlib to explore revenue patterns across time, category, region, and product.

### Monthly Revenue Trend



Tracks total revenue by month, highlighting seasonal peaks and troughs across the two-year period.

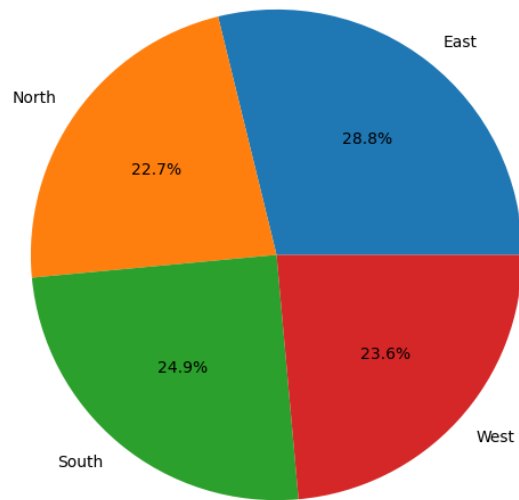
### Revenue by Category



Compares total revenue contribution across product categories such as Sports, Electronics, and Books.

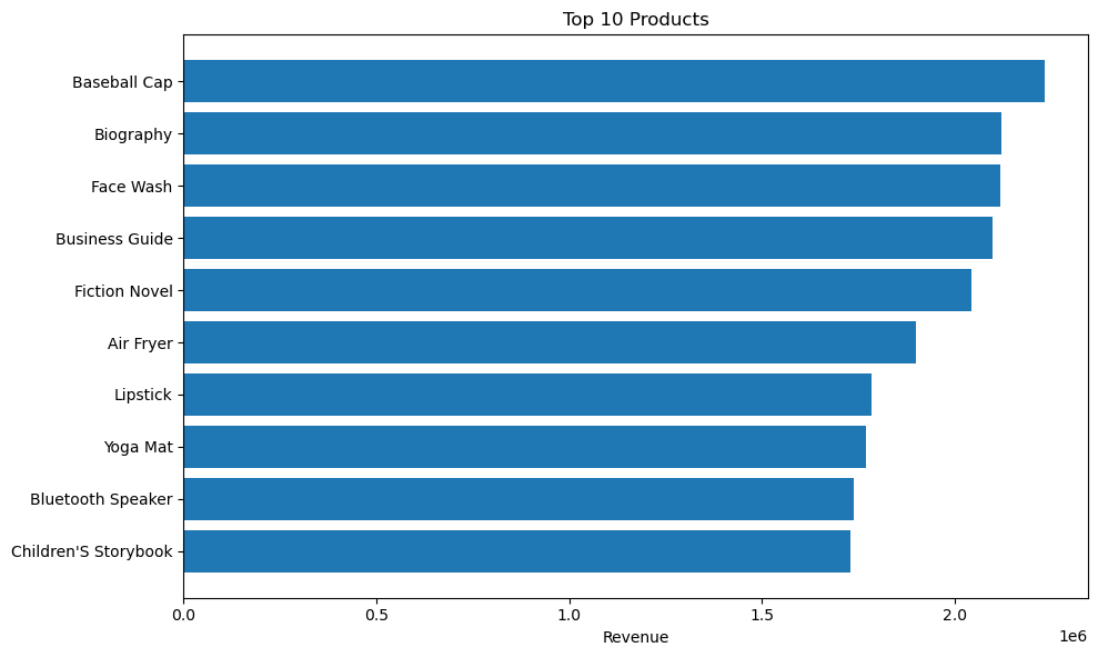
### Revenue by Region

Revenue by Region



Shows the percentage share of revenue contributed by each of the four sales regions.

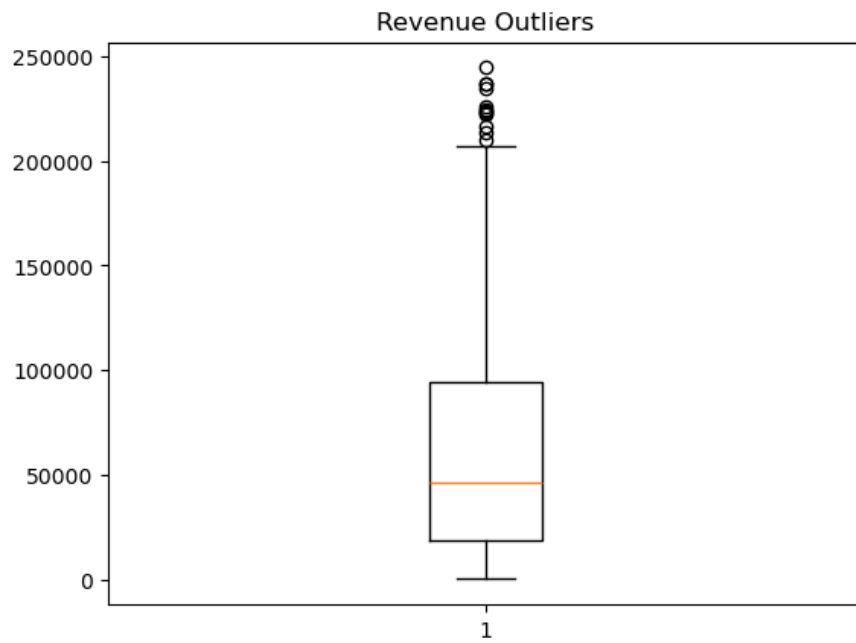
## Top 10 Products by Revenue



Ranks the ten highest revenue-generating products across the entire dataset.

## Revenue Distribution & Outliers





*A boxplot used during data validation to detect unusually high or low revenue values.*

## 7. Key Insights

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- Revenue is distributed across four regions with no single region overwhelmingly dominant, indicating a geographically balanced customer base.
- A small set of top products account for a disproportionately large share of total revenue, consistent with a typical Pareto (80/20) sales pattern.
- Monthly revenue shows visible fluctuation across the year, suggesting seasonal demand that could inform inventory and marketing planning.
- Category-level revenue comparison highlights which product lines are the strongest contributors versus underperforming categories.
- Data quality issues (mixed date formats, currency-formatted prices, inconsistent text casing) were significant enough to require a dedicated cleaning stage before any analysis could be trusted.

## 8. Conclusion

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This project demonstrates a full analytics workflow: ingesting a raw, inconsistent retail sales export, systematically cleaning and validating it, engineering time and revenue-based features, and producing clear visual insights into sales performance. The cleaned dataset and derived metrics provide a reliable foundation for further analysis or dashboarding (e.g. in Power BI).